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Software Requirements Specification

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

Field	This example
Team name	5 Idiots
Project title	Pantry Pal Application— A grocery expiration and food waste management application
SDLC chosen in M1 + reason	Agile — develop features in short iterations, test with users, and continuously improve the system based on feedback.
Date / version	November / v1.0.1
GitHub repo	https://github.com/Dora7258/Pantry-Pal.git

1. Introduction

1.1 Purpose and scope

The purpose of this application is to help users manage their groceries and reduce food waste by keeping track of food items and their expiration dates. Users can add grocery items, record their expiration dates, and receive notifications before items expire. The application also helps users identify which groceries should be used first, so they can avoid forgetting food in the refrigerator or pantry, reduce unnecessary spending, and minimize food waste. The initial version focuses on grocery tracking, expiration alerts, and inventory management, with additional features such as recipe suggestions and food waste tracking added later.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Household member	primary user	View and manage shared grocery items and know which items should be used first	Accidentally deleting or changing another person's grocery information
Application administrator	institution or supplier	Manage the application, monitor system performance, and maintain user data	Data loss, system errors, or unauthorized access to user information

2. Overall description

2.1 Product context — what it does / does not

The Grocery Management Application helps users keep track of their groceries and expiration dates. Users can add, edit, and delete grocery items, record expiration dates, receive alerts before items expire, and view items that need to be used soon. The MVP focuses on preventing users from forgetting about groceries and reducing food waste. Additional advanced features are outside the scope of the MVP.

IN SCOPE (MVP this term)		OUT OF SCOPE (explicit promise)
FR-1 — Create / edit / delete account and upgrade to Premium **FR-2** — Add / edit / delete/ mark as used grocery items **FR-3** — Automatically calculate and display the estimated expiration date based on the grocery category and purchased date **FR-4** — Expiration date reminders / notifications **FR-5** — Display active items **FR-6** — Show items that expire soon and already expired **FR-7** — Premium receipt or items scanning **FR-8** — Automatic category detection **FR-9** —- Premium custom reminder settings **FR - 10** —- Expired grocery history(Free for 7 days, unlimited for premium) **FR-11** — Premium usage summary (daily, weekly, and monthly)		Group chat or messaging Online grocery shopping or purchasing directly through the app. Automatic grocery delivery services. Nutrition, calorie, or health tracking. Automatic detection of whether food is spoiled or safe to eat. Integration with supermarkets or grocery stores.

2.2 Assumptions and constraints

- **Assumption:** Users have a smartphone and can access the application.
- **Assumption:** Users will enter accurate grocery information, especially the purchased date and expiration date.
- **Constraint:** The application supports basic grocery information such as item name, quantity, purchase date, and expiration date.

- **Constraint:** The system will not automatically determine whether food is spoiled; it only tracks the information provided by the user.
- **Constraint:** The project must be completed within the assigned development period and available team resources.

3. Functional requirements

As a `<role>`, I want `<goal>`, so that `<benefit>`.

FR-1 — Create / edit / delete account/ upgrade to Premium

User story: As a user, I want to create, edit, delete my account, and subscribe to Premium, so that I can manage my account and access Premium features.

Pain this traces to: Users need to manage their account information and may want to access additional features through a Premium subscription.

1. Given the user enters the required account information, when they create the account, the account is saved and can be used to access the application.
2. Given an existing account, when the user edits or deletes their account information, the updated information is saved and displayed.
3. Given an existing account, when the user chooses to subscribe to Premium, the account's subscription status is updated to Premium and Premium features become available.
4. Given a free account, when the user chooses to upgrade to premium subscription, the account status is changed to premium and premium features are available.

FR-2 - Manage grocery items

User story: As a user, I want to add, edit, delete, and mark grocery items as used, so that I can keep my grocery inventory up to date.

Pain this traces to: Users may forget what groceries they have purchased and when they expire. Manually managing many grocery items can also take too much time.

1. The system allows the user to add grocery items manually, including the grocery name, category, quantity, and purchased date.
2. The system allows the user to edit or delete existing grocery items.
3. The system allows the user to mark a grocery item as used, removing it from the active inventory while keeping its status as used.

FR-3 — Automatically calculate and display the estimated expiration date based on the grocery category and purchased date

User story: As a user, I want Pantry Pal to estimate the expiration date of my groceries based on their category and purchased date, and quantity so that I do not have to calculate it myself.

Pain this traces to: Users may find it difficult to remember or estimate expiration dates for different types of groceries.

1. The system identifies the selected grocery category and purchased date.
2. The system calculates an estimated expiration date using the predefined shelf-life period for the selected category.
3. The system displays the estimated expiration date and allows the user to edit it before saving.

FR-4 — Expiration reminders / notifications

User story: As a user, I want to receive reminders before my groceries expire, so that I can use them in time.

Pain this traces to: Users may forget upcoming expiration dates and only notice that food has expired after it is too late.

1. The system identifies grocery items that are approaching their expiration dates before 3 days.(Default for free users, and Premium users use the period set in FR-9)
2. The system sends a notification according to the user's selected reminder setting.
3. The notification identifies the grocery item and its expiration date.

FR-5 — Display active items

User story: As a user, I want to see my active grocery items in one place, so that I can keep track of what I currently have.

Pain this traces to: Users may forget which groceries they currently have, especially when items are stored in different places.

1. The system displays all grocery items that are currently active.
2. The system displays relevant information such as the item name, category, quantity, and expiration date.
3. The system removes an item from the active inventory when the user deletes it or marks it as used.

FR-6 — Show items that expire soon and already expired

User story: As a user, I want to see which groceries are expiring soon or have already expired, so that I can prioritize which items need attention.

Pain this traces to: Users may use newer groceries first while older groceries remain unused and eventually expire.

1. The system identifies groceries that are approaching their expiration dates.(within 3 days for normal user, or within premium user's custom reminder period)
2. Items that are expiring soon or already expired are displayed according to their expiration status.
3. The user can distinguish between **expiring-soon** and **expired** items.

FR-7 — Premium receipt or items scanning

User story: As a Premium user, I want to scan a grocery receipt or item, so that I can add multiple grocery items without entering them one by one.

Pain this traces to: Manually adding many grocery items after shopping can take too much time.

1. The system allows Premium users to scan a grocery item or receipt using the device camera.
2. The system extracts and displays the detected grocery items for the user to review and confirm.
3. The system adds the confirmed grocery items to the grocery inventory.

FR-8 — Automatic category detection

User story: As a premium user, I want the system to automatically detect grocery categories, so that I do not have to categorize each grocery item manually.

Pain this traces to: Categorizing many grocery items individually can take additional time after shopping.

1. The system identifies a suggested category for each detected grocery item.
2. The system displays the suggested category for the user to review and confirm.
3. The system saves the confirmed category with the grocery item.

FR-9 — Premium custom reminder settings

User story: As a Premium user, I want to customize when I receive expiration reminders, so that I can choose a reminder time that suits my needs.

Pain this traces to: Different users may need different amounts of time to use their groceries before they expire.

1. The system allows Premium users to select a preferred reminder period before expiration.
2. The system saves the user's selected reminder setting.
3. The system sends expiration notifications according to the selected reminder setting.

FR-10 — Expired grocery history

User story: As a user, I want to view my expired grocery history, so that I can review groceries that have expired in the past.

Pain this traces to: Users may want to review previously expired groceries to understand their food-waste patterns.

1. The system allows Free users to view expired grocery history from the **past 7 days**.
2. The system allows Premium users to view their expired grocery history **without the 7-day limit**.
3. The system displays information such as the grocery name, expiration date, and expired date.

FR-11 — Premium usage summary

User story: As a Premium user, I want to view a daily, weekly, and monthly summary of the groceries I used and wasted, so that I can understand my food-waste habits and use my groceries before they expire.

Pain this traces to: Users may want to understand how much food they use on time and how much they waste, but a list of past items does not clearly show their food-waste patterns.

1. The system allows only Premium users to view the usage summary in daily, weekly, and monthly views. Free users see a message that this is a Premium feature, with an **Upgrade to Premium** option.
2. The system displays the number of **expired items**, **items used before expiring**, and **items used after expiring** for the selected period, based on the date each item was marked as used. It shows these counts in a bar chart, or a message when there is no data.
3. The system resets the three counts to **0 at the end of each month** and displays the next reset date.

4. Non-functional requirements

Forbidden words unless quantified: fast, easy, user-friendly, smart, secure. NFR-(security/privacy) is required for M2 — every SRS must have at least one.

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	The system shall save account and grocery-item information within 2 seconds after valid submission and process a valid receipt scan within 5 seconds.	Measure the saving and receipt-processing time over 3 trials.
NFR-2	usability	A first-time Free user shall be able to add a grocery item manually within 60 seconds without assistance, and a first-time Premium user shall be able to scan and confirm a receipt within 60 seconds without assistance.	Conduct separate timed tests with first-time Free and Premium users.
NFR-3	reliability	The system shall retain each user's account information, grocery information, subscription status, receipt-scan usage, reminder settings, and expired-history information after the application and usage-summary information (including the date each item was marked as used) after the application is closed and reopened.	Add information, perform a receipt scan, close the application, reopen it, and verify that the data remains.
NFR-4	security / privacy	A user shall only be able to access and modify their own account, grocery inventory, view their own subscription information, expired-history information, and usage summary information.	Log in as one user and attempt to access another user's information.
NFR-5	availability	The system shall be available at least 99% of the time, excluding planned maintenance.	Monitor system availability over a defined testing period and calculate the percentage of uptime.
NFR-6	compatibility	The application shall function correctly on supported Android and iOS devices and different supported screen sizes.	Test the application on multiple supported devices and screen sizes.
NFR-7	Notification reliability	The system shall deliver expiration notifications according to the default reminder setting for Free users and the selected custom reminder setting for Premium users.	Test notifications using both Free and Premium accounts with their respective reminder settings
NFR-8	maintainability	The system shall use a modular structure so that account, grocery, scanning, and notification functions can be maintained independently.	Review the system structure and verify that the major functions are separated into appropriate modules.
NFR-9	data accuracy	The system shall correctly calculate the estimated expiration date based on the grocery category and purchased date, and Premium automatic category detection shall correctly display the detected category for user confirmation.	Enter test grocery categories and purchase dates, compare the estimated expiration dates with expected results, and test Premium category detection using sample receipts.

ID	Kind	Statement (testable)	How we will check
NFR-10	error handling	The system shall display a clear error message when required or invalid account/grocery information is entered.	Enter missing and invalid information and verify that an appropriate error message is displayed.
NFR - 11	Data Accuracy	The system shall count each grocery item in exactly one category (expired, used before expiring, or used after expiring) in the usage summary for the selected period, and shall reset these counts to 0 at the end of each month.	Mark test items as used before and after their expiration dates and let others expire, compare the daily, weekly, and monthly counts with the expected results, then simulate the end of the month and verify the counts are 0.

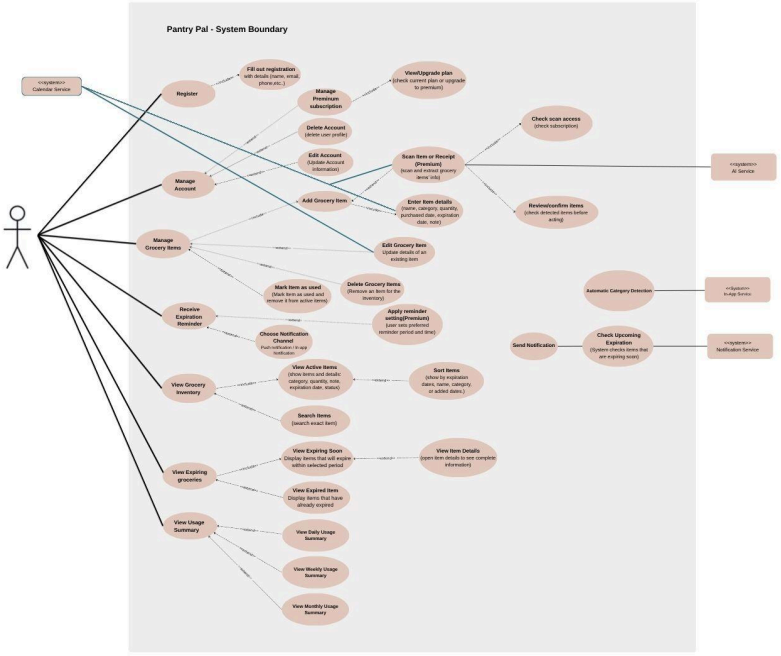
5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1 Register	Household Member	FR-1	The household members create their account
UC-2 Manage account	Household Member	FR-1	The household members create, edit, or delete their account, save or remove the account information accordingly.
UC-3 Manage grocery items	Household Member	FR-2,FR-3	The household member manually adds, edits, deletes, or marks a grocery item as used, and the inventory is updated accordingly. The system also estimates the expiration date from the grocery category and purchased date.
UC-4 Scan Item or Receipt	Household Member	FR-7,FR-2, FR-8	The household member scans a receipt or item to add multiple grocery items instead of entering them manually.
UC-5 Check Scan Access	Household Member	FR-7	The system checks the user's subscription and Premium users may continue scanning, while Free users are shown an upgrade option.
UC-6 Review / Confirm Scanned Items	Household Member	FR-8, FR-2	The system displays the detected grocery items, and the household member reviews and confirms them before adding them to the inventory.
UC-7 Automatic Category Detection	Household Member	FR-8	The system automatically identifies and assigns a category to a grocery item.
UC-8 Receive Expiration Reminder	Household Member	FR-4	The system checks upcoming expiration dates and sends a reminder to the household member.

Use case	Actor	Related FR	Brief success story
UC-9 Customize Reminder Settings	Household Member	FR-9	A premium user selects their preferred reminder timing and notification settings.
UC-10 View Grocery Inventory	Household Member	FR-5	The household member opens the inventory and sees all currently active groceries, including their names and expiration dates.
UC-11 View Expiring / Expired Items	Household Member	FR-6	The household member views groceries that are expiring soon or have already expired and identifies which items need attention.
UC-12 View Expired Grocery History	Household Member	FR-10	The system shows expired grocery history according to the user's subscription: 7 days for free users and unlimited for premium users during the subscribed period.
UC-13 Upgrade to Premium	Household Member	FR-1	A free household member upgrades their account to Premium, and the system changes the subscription status to enable Premium features (receipt scanning, custom reminder settings, and unlimited expired grocery history for subscribed period and usage summary).
UC-14 View Usage Summary	Household Member	FR-11	A Premium household member opens the usage summary and selects a daily, weekly, or monthly view to see the number of expired items, items used before expiring, and items used after expiring. Free members are shown an upgrade option instead.

5.2 Use-case diagram



6. Out of scope

1. Online grocery shopping or purchasing directly through the app.
2. Automatic grocery delivery services.
3. Nutrition, calorie, or health tracking.
4. Automatic detection of whether food is spoiled or safe to eat.
5. Integration with supermarkets or grocery stores.
6. Sharing or messaging with other users.

7. Traceability matrix (Golden Thread)

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Users need to manage their personal information, and account securely, and basic services take time to process managing items.	**FR-1** Create, edit, delete account, and upgrade to Premium	• User account entity • Registration form • Authentication • Account management • Premium Subscription	• Registration page • Login page • Account Settings	TC-01: Register, log in, edit account details and verify Premium access.
Users forget what groceries they have and may waste food or consume expired food.	**FR-2** Add, edit, and delete grocery items and mark them as used.	• Grocery item entity • Expiration date field • CRUD operations • Item status	• Pantry Dashboard	TC-02: Add, edit, delete and mark grocery items as used.

		(active/used) (Ch 5 & 7)	• Add/Edit Grocery page	
Users may find it difficult to remember or estimate expiration dates for different types of groceries.	**FR-3** Automatically calculate and display the estimated expiration date based on the grocery category and purchased date.	• Category-based shelf-life period • Calculate expiration date using purchase date • Allow date editing	• Add/Edit Grocery page	TC-03: Select a grocery category and purchase date; verify the estimated expiration date.
Users may forget about groceries until they expire, which can lead to food waste and possible health risks from consuming expired food.	**FR-4** Send expiration alerts/ notifications.	• Expiration-date checking • Reminder settings • Notification service	• Reminders page • Notification interface	TC-04: Check whether expiration reminders follow the configured reminder period.
Users may not remember what groceries they currently have at home.	**FR-5** Display all active grocery items with details.	• Query active items • Display grocery name and expiration date • Display item status	• Pantry Dashboard • Active Items section	TC-05: Add grocery items and verify that all active items appear with correct details.
Users may overlook groceries that are about to expire or have already expired.	**FR-6** Show items that are expiring soon and already expired.	• Filter items by expiration date • Identify expiring-soon items • Identify expired items • Sort by expiration date	• Pantry Dashboard • Expiring Soon section • Expired Items section	TC-06: Test different expiration dates and verify the correct item status.
Manually adding many grocery items after shopping can take too much time.	**FR- 7** Premium receipt scanning	• Camera scan •Premium-only access check •Review and confirm screen	• Premium Features page • Scan Receipt page	TC-07: Scan a sample receipt and verify that detected items are displayed for confirmation
Categorizing many grocery items individually can take additional time after shopping	**FR-8** Automatic category detection	• Automatic category detection • Suggested category per item • Confirm/edit category	• Scan Review page • Add/Edit Grocery page	TC-08: Check whether detected grocery items receive appropriate category suggestions.
Different users may need different amounts of time to use their groceries before they expire	**FR-9** Premium custom reminder settings	•Reminder-period setting •Saved user preference •Notification service	• Premium Features page • Reminder Settings page	TC-09: Change the reminder period and verify that the selected setting is saved.
Users may want to review previously expired groceries to understand their food-waste patterns.	**FR-10** Expired grocery history(Free: past 7 days, Premium:Unlimited)	• Expired-item history query • 7-day limit for Free users	• Grocery History page • Premium History section	TC-10: Check expired-item history access for Free and Premium accounts.

		• Unlimited history for Premium users		
Users may want to understand how much food they use on time and how much they waste	**FR-11** Premium usage summary (daily, weekly and monthly).	• Usage statistics calculation • Daily, weekly and monthly summaries • Bar chart visualization • Monthly count reset	• Usage Summary page • Premium Features page	TC-11: Verify daily, weekly and monthly usage counts using sample grocery records.

8. AI usage log

Date	*Tool / model*	*What we asked*	*What AI produced*	*What we changed / verified*	*What we learned*
22 Aug 2026 *29 Aug 2026*	*Chat GPT/ Claude/ Perplexity*	*Help us to modify the requirements and use cases for a Pantry Pal System, including functional requirements, non-functional requirements.*	*AI suggested FRs, NFRs, and a detailed for managing groceries, expiration dates, reminders, and scanning.*	*We reviewed the suggestions, changed the actor to Household Member, corrected the FR and use-case relationships, removed unnecessary details, and verified that manual entry belongs to grocery management*	*We learned how to distinguish FRs from NFRs, connect FRs to use cases, and use include/extend relationships appropriately.*
18 Sep 2026 *22 Sep 2026*	*Chat GPT/ Claude/ Gemini, Reddy AI*	*To check our use case diagram and requirements*	*It showed the things that are not matched and errors in the file.*	*We reviewed and checked whether it is the correct suggestion or not and fixed it if needed*	
10 Oct 2026	*Claude AI ChatGPT*	*To build our application*	*It gave us the file which features do not match with our FRs.*	*We reviewed and changed it to get the version which matches with FRS..*	
2 Nov 2026	*Claude AI/ ChatGPT*	*To upgrade our application*	*It gave us the version which is the same as the old version.*	*We reviewed and changed to get which is to add more features.*	

9. Self-Check

- **Every FR is a user story with a benefit** — all 10 **FRs** include a “so that” clause connected to a real grocery-management problem.
- **Every FR has acceptance criteria you could actually test** — each FR has clear and testable acceptance criteria.
- **No FR is vague** — the FRs use specific actions such as create, add, edit, delete, display, and notify.
- **NFRs exist, including security / privacy** — NFRs cover performance, usability, reliability, security/privacy, compatibility, data accuracy, and error handling.
- **Every FR traces back to a stated problem** — each FR includes a **“Pain this traces to”** statement explaining the problem it addresses.
- **Scope and out-of-scope are stated** — the project scope and features that are not included are clearly defined.
- **Use-case list matches the FR list** — there are **13 use cases**, covering all 11 FRs, with **Household Member** as the main actor.
- **AI log is honest** — the AI usage log shows what we asked AI to help with and what we reviewed and changed.